

CRM SALES OPPORTUNITY FORECASTING

Machine Learning Classification Assignment

Submission PDF

GitHub Repository: https://github.com/mohitparnami18/ML-Assignment2-CRM_Sales_Forecasting_Assignment

Live Streamlit Community Cloud App: <https://ml-crmsalesforecastingassignment.streamlit.app/>

Models: Logistic Regression, Decision Tree, KNN, Gaussian Naive Bayes, Random Forest, Gradient Boosting

Evaluation: Accuracy, AUC, Precision, Recall, F1 Score and MCC

This document follows the mandatory submission order and includes the README content and BITS Virtual Lab execution evidence.

1. GitHub Repository Link

Repository: https://github.com/mohitparnami18/ML-Assignment2-CRM_Sales_Forecasting_Assignment

The repository contains the complete source code, requirements.txt, README.md, test_data.csv, saved model files and supporting training/feature-engineering scripts.

Repository entry point: app.py

Streamlit main file path: app.py

2. Live Streamlit App Link

Live application: <https://ml-crmsalesforecastingassignment.streamlit.app/>

The deployed application provides test-data upload, six-model selection, a complete model comparison table, evaluation metrics, confusion matrix, classification report and prediction results.

Deployment status: Public Streamlit Community Cloud URL supplied by the student.

3. BITS Virtual Lab Execution Screenshot

The following screenshot shows the CRM Sales Forecasting application running in the BITS Virtual Lab environment. It demonstrates the test dataset, six-model comparison and evaluation results.

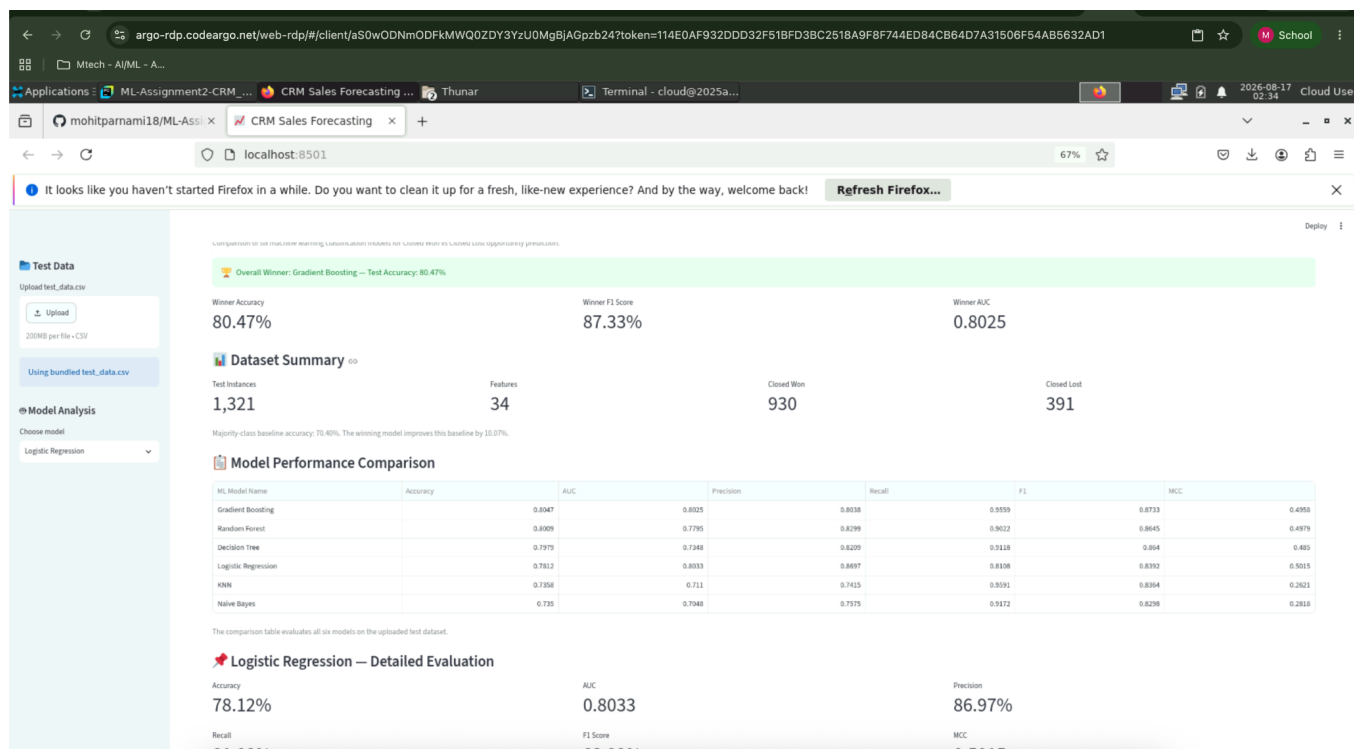


Figure 1. BITS Virtual Lab execution showing the deployed-style Streamlit interface and model comparison.

4. GitHub README Content

a. Problem Statement

The objective of this project is to predict whether a closed CRM sales opportunity will be **Closed Won** or **Closed Lost** using customer/account characteristics, sales-source information, owner information, historical account activity and historical sales performance.

This is formulated as a binary classification problem: **Closed Won = 1** and **Closed Lost = 0**.

b. Dataset Description

Source: Kaggle — Sales CRM Data. The dataset contains CRM entities including Opportunity, Account, Lead, Contact, Order and User. The Opportunity table contains approximately 16,000 records and the Account table approximately 10,000 records. The final supervised dataset contains 6,604 closed opportunities and 34 model features.

Target: Closed Won = 1; Closed Lost = 0.

Test data: 1,321 instances, comprising 930 Closed Won and 391 Closed Lost opportunities (70.40% win rate).

Leakage prevention: stage_name, probability, close_date and days_to_close are excluded because they can reveal the final outcome. Historical performance features are calculated using information available before the current opportunity.

Split: 60% training, 20% validation and 20% test using a chronological strategy.

c. Repository and Application Links

GitHub: https://github.com/mohitparnami18/ML-Assignment2-CRM_Sales_Forecasting_Assignment

Streamlit: <https://ml-crmsalesforecastingassignment.streamlit.app/>

d. Models Used

- Logistic Regression
- Decision Tree Classifier
- K-Nearest Neighbor Classifier
- Gaussian Naive Bayes
- Random Forest
- Gradient Boosting

The assignment wording refers to six models while explicitly listing five. Gradient Boosting is included as the sixth model.

Model Performance Comparison

ML Model Name	Accuracy	AUC	Precision	Recall	F1	MCC
Gradient Boosting	0.8047	0.8025	0.8038	0.9559	0.8733	0.4958
Random Forest	0.8009	0.7803	0.8279	0.9054	0.8649	0.4965
Decision Tree	0.7979	0.7348	0.8209	0.9118	0.8640	0.4850
Logistic Regression	0.7812	0.8033	0.8697	0.8108	0.8392	0.5015
KNN	0.7358	0.7110	0.7415	0.9591	0.8364	0.2621
Naive Bayes	0.7350	0.7048	0.7575	0.9172	0.8298	0.2818

Observations on Model Performance

Logistic Regression: 78.12% accuracy and the highest AUC (0.8033), precision (86.97%) and MCC (0.5015). It provides strong balanced performance and is useful when precision is important.

Decision Tree: 79.79% accuracy and 86.40% F1. Its 91.18% recall identifies most Closed Won opportunities, although its AUC is lower than the stronger ensemble models.

KNN: 73.58% accuracy and 0.7110 AUC. It has very high recall (95.91%) but comparatively weak MCC (0.2621), indicating weaker overall class separation.

Naive Bayes: 73.50% accuracy and 0.7048 AUC. Its high recall (91.72%) is offset by comparatively low MCC (0.2818).

Random Forest: 80.09% accuracy and 86.49% F1, providing a strong balance between precision and recall.

Gradient Boosting: Highest accuracy (80.47%) and F1 (87.33%), with 95.59% recall and 0.8025 AUC. It provides the strongest overall test performance.

Overall Winner: Gradient Boosting based primarily on test accuracy and F1 score. Logistic Regression nevertheless has the highest AUC, precision and MCC, so the preferred model can depend on the business objective.

Majority-class baseline: 70.40% accuracy. Gradient Boosting improves this baseline by approximately 10.07 percentage points.

5. Streamlit Application Features and Execution Evidence

- Dataset upload option for test_data.csv
- Model selection dropdown covering all six models
- Complete comparison table across all six models
- Accuracy, AUC, Precision, Recall, F1 and MCC display
- Confusion matrix
- Classification report
- Prediction results and CSV download

The second screenshot below shows the detailed evaluation view in the BITS Virtual Lab execution environment.

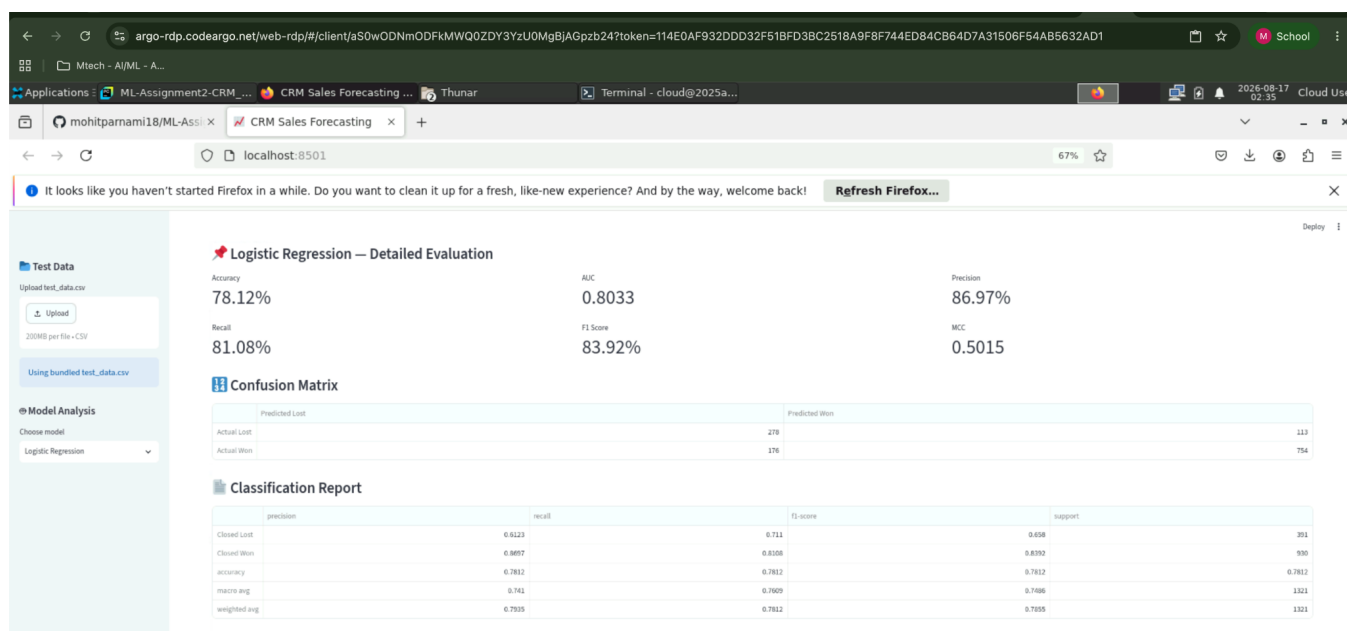


Figure 2. Detailed Logistic Regression evaluation, confusion matrix and classification report.

6. Final Submission Checklist

Requirement	Status
GitHub repository link	Completed
Live Streamlit Community Cloud link	Completed
BITS Virtual Lab screenshot	Included
README content included in PDF	Included
Six classification models	Implemented
Six required evaluation metrics	Included
Test data CSV	Included in repository
Streamlit dataset upload	Implemented
Streamlit model selection	Implemented
Streamlit evaluation metrics	Implemented
Confusion matrix / classification report	Implemented

Submission links:

GitHub: https://github.com/mohitparnami18/ML-Assignment2-CRM_Sales_Forecasting_Assignment

Streamlit Community Cloud: <https://ml-crmsalesforecastingassignment.streamlit.app/>

End of submission document.